

Analytics and Applications [AA] WS 2025/26

Master of Science
BM Information Systems II / BM Artificial Intelligence and Visual Analytics III
Faculty of Management, Economics, and Social Sciences
Department of Information Systems for Sustainable Society
University of Cologne
Version 7 - December 10, 2025

Instructor Prof. Dr. Wolfgang Ketter **Term** WS 2025/26
TA Julius Kuhmann **Class Time** Wed, 14.00-17.30h
Website www.is3.uni-koeln.de and ILIAS **Class Location** 136 Kleiner Hörsaal XXXI OR
211 Hörsaal H115 (see ILIAS)

Welcome to **Analytics and Applications [AA]**. This course and the accompanying reading materials aim to provide the knowledge and skills required for data analytics using information systems that drive business success.

Business Analytics is the use of data-driven decision-making. Companies, governments, and other organizations now collect and have access to large amounts of data about suppliers, clients, employees, citizens, transactions, etc. Data Mining and predictive analytics provide a powerful toolkit for detecting actionable patterns in data and generating predictions. These methods are used in many industries: Mobile companies use their customer database to predict customer churn or to personalize SMS messages to improve customer service; Financial institutions use past loan data to predict defaulting chances for loan applicants; Charities use data from a campaign in one location to target the right people in another location; Politicians use databases of supporters to segment and best target each audience; Movie rental and e-Commerce websites provide recommendations based on users' online behavior; Renewable energy providers use weather forecasts to predict their electricity generation to better trade in different types of markets.

Most business analytics applications are geared toward benefiting the company, often at the expense of the individual, community, and society. This course focuses on human and socially-responsible business analytics, especially with a focus on sustainability.

In this course, we will work with real business problems and real data. We will examine the types of questions that data mining can answer and will develop a variety of data-driven tools to answer these questions. The emphasis is on understanding the concepts and logic behind a wide set of data mining techniques and their relation to specific business analytics situations. The course is not about mastering the theoretical underpinnings of the techniques. The gained knowledge will be applied to a business analytics project that encapsulates the learnings, while is not expected to master the theoretical underpinnings of the techniques.

You will learn about the process of data analytics. You will learn to identify problems, define the structure of an information system, evaluate competing solutions for a business problem, and you will start to “speak business and analytics”. You will gain an insight into how important these activities are in creating information systems that are truly aligned with business needs. Throughout the course, you will also learn about selected topics in sustainability, especially in terms of renewable electricity.

Some of the most important thoughts that we will present throughout the lecture are summarized in this reader and the references therein. This material is designed to help you find additional references, and to help you recapture what was taught in the lectures. It is not designed, however, to serve as an exclusive source of the exam preparation material. In our lectures and workshops, we will present facts, and examples, and teach you skills that are not on the following pages, and we reserve the right to ask you about them during the final exam of your assignments. Please make sure you take advantage of all modes of learning that we offer in this course!

Cologne, December 10, 2025

Prof. Dr. Wolfgang Ketter, Julius Kuhmann
Department of Information Systems for Sustainable Society

Class Schedule

#	Title	Topics	Reading ¹
#1 Oct 15	(L) Kick-Off (W) Workshop 1	Intro to course contents & objectives Admin & Setup	Ch. 1
#2 Oct 22	(L) Intro Data Science (W) Workshop 2	Core ideas and concepts in data mining tasks Recap & Coding	Ch. 2
Oct 27	Deadline KLIPS Examination Sign-Up	See Section 1.2	-
#3 Oct 29	(L) Regression 1/2 (L) Regression 2/2	Linear Regression & Polynomial Features Cross-Validation & Regularization	Ch. 6
Nov 05 latest	Start Project Assignment	See Section 2	-
#4 Nov 05	(L) Classification 1/2 (W) Workshop 3	Separating Boundary Classification (SVM) Recap & Coding	Ch. 6
#5 Nov 12	(W) Workshop 4 (W) Workshop 5	Recap & Coding Recap & Coding	-
#6 Nov 19	(L) Naive Bayes (L) Classification 2/2	Probability Recap, Naive Bayes Classifier Probabilistic Classification	Ch. 8 & 10 Ch. 9
#7 Nov 26	(W) Workshop 6 (W) Workshop 7	Recap & Coding Recap & Coding	-
Nov 26 23:59	Deadline Milestone 1	See Section 2.2	-
#8 Dec 03	(L) Ensemble Methods (L) Neural Networks	Trees, Bagging, Boosting Intro to Artificial Neural Networks (NN)	Ch. 13 Ch. 11
#9 Dec 10	(W) Workshop 8 (W) Workshop 9	Recap & Coding Recap & Coding	-
#10 Dec 17	(L) Unsupervised Learning 1/2 (W) Workshop 10	K-Mean, K-means++, Hierarchical clustering Recap & Coding	Ch. 15
-	Christmas / New Years	Happy Holidays!	-
#11 Jan 07	(L) Unsupervised Learning 2/2 (L) Recommender Systems	Dimensionality Reduction, Fuzzy Clustering Association Rules & Collaborative Filtering	Ch. 15
Jan 07 23:59	Deadline Milestone 2	See Section 2.2	-
#12 Jan 14	(W) Workshop 11 (W) Workshop 12	Recap & Coding Recap & Coding	-
#13 Jan 21	(L) Time Series (L) Text Mining	Time Series Analysis and Forecasting Natural Language Processing (NLP)	Ch. 16-19 Ch. 20
#14 Jan 28	(W) Workshop 13 (W) Workshop 14	Recap & Coding Recap & Coding	-
#15 Feb 04	(L/W) Q&A Session	General Q&A session at the end	-
Feb 08 23:59	Deadline Project Assignment	see Section 2.2	-
Feb 20	Exam 1	16:00 - 17:30, 105 Hörsaal A2	-
Mar 20	Exam 2 (only in case Exam 1 was not passed; see Section 1.2)	10:00 - 11:30, 105 Hörsaal C	-

Table 1. Tentative Schedule, (L): Lecture Slot, (W): Workshop Slot

¹: Recommended Reading: Shmueli et al. (2019)

Course Requirements and Resources

1 Programming Software

An important feature of this course is hands-on learning using data mining and data visualization software. For data mining, you will use the free, open-source Python programming software. We recommend using the Anaconda distribution of Python. There are two options:

- **Option 1** (probably the easiest): Install the full Anaconda version, which comes with a lot of pre-installed packages (incl. Jupyter Notebook) and a package manager graphical user interface (GUI). It is available for download **here**.
- **Option 2** (for slightly more advanced users and if you are short on disk space): Install the much lighter MiniConda distribution, which essentially is just the Conda package manager and Python. From there, you can simply install Python packages as required via the Terminal (Mac) or the Command Prompt (Windows). Miniconda is available for download **here**.

Please refer to the Git repository (see below) for detailed instructions on the software set-up, which should be completed before the first workshop session. Every student is responsible for bringing a laptop to each workshop session. Please note that given the typical size of the cohort we do not have capacity to provide one-on-one tech support should you encounter software issues. Luckily Python has excellent community support and you are almost guaranteed to find a solution to your specific problem in a matter of a few Google queries.

2 Online Git Repository

We operate an official git repository for AA. We will use this repo as the central point for sharing workshop materials, codes, data, and instructions with you over the course of this semester. This repo contains the code and (some) data for workshops.

The git repository is hosted on GitHub. The link is the following:

- **Git Address:** https://github.com/IS3UniCologne/AA_2025.git

You should check this repository regularly to view and/or download newly released or updated files. If you have experience using git, you may also wish to clone the repository on your local machine for convenience.

3 ILIAS Platform

ILIAS will be used for sharing all lecture-related content, including copies of lecture slides. Lecture slides will be uploaded after each lecture. We strongly recommend to attend each lecture live as this will facilitate the best learning journey.

You can access ILIAS using the following link and password:

- **Course Folder Name:** Analytics and Application [MAIN FOLDER] (can be accessed **here**)
- **Password:** AA_2025!

4 Textbooks

Core Recommended Reading

- **Data Mining for Business Analytics: Concepts, Techniques and Applications in Python** by Galit Shmueli, Peter C. Bruce, Peter Gedeck & Nitin R. Patel, Wiley, 2019, ISBN: 978-1-119-54986-4.

Further Reading

- **Python Data Science Handbook** by Jake VanderPlas, O'Reilly, 2016. [available as free .pdf from [here](#)]: A theory-light methods-focused reference that covers the most important Data Science Python libraries in good detail.
- **An Introduction to Statistical Learning** by Gareth James, Daniela Witten, Trevor Hastie & Robert Tibshirani, Springer, 2021. [available as free .pdf [here](#)]: A very accessible introduction into core machine learning concepts and their applications.
- **Pattern Recognition and Machine Learning** by Christopher Bishop, Springer, 2006. [available as free .pdf from [here](#)]: A comprehensive introduction to the fields of pattern recognition and machine learning that assumes no prior knowledge on the topics.

Assessment Guidelines

1 General Guidelines

1.1 Assessment Components

Assessment of individual performance in **AA** will be based on two main components: (1) a **project assignment** (see Section 2 for details) conducted over the second and third part of this course and (2) a **written exam** (see Section 3 for details) taking place at the end of the semester. For the project assignment, you will be working in teams and required to deliver a project report. All material covered as part of this course (lectures and workshops) will be exam-relevant, and your understanding of these topics will be tested in the final exam. Please refer to the table below for weights and deadlines for the individual assessment components.

Assignment	Percentage of Grade	Date/Deadline
Project Assignment	33%	Feb 08 23:59
Exam	67%	see Class Schedule
Total	100%	

Table 1. Grading Scheme & Dates/Deadlines

Students need to pass both assessment components (Project Assignment and Exam) individually in order to successfully complete this course. As per University-wide regulations this means that you will need to achieve a grade of 4.0 or better in both examination components.

Note: Project Assignment grades **cannot** be carried over into the next academic year. More concretely: if you do not pass this course, you will have to re-do the Project Assignment next year.

Note: Those students who passed the bachelor's course "Introduction to Data Science and Machine Learning (DSML)" **cannot** take over their grades from that course to pass the course Analytics and Applications.

1.2 KLIPS Registration

It is mandatory to register for the Examination Object in KLIPS **before Oct 27, 23:59** to be able to pass this course. At the same time, if you decide not to take this course, make sure to de-register before the same date as a later decision not to take this course is only possible via *Rücktritt mit anerkanntem Grund* (see <https://wiso.uni-koeln.de/de/fakultaet/dekanat/zentrale-fakultaetsverwaltung/pruefungsamt/po-2015-po-2021/rund-um-pruefungen/nachtraeglicher-ruecktritt>).

2 Project Assignment

The project assignment runs in parallel to the lectures. Deadline is **February 08th, 23:59**, submission will be through ILIAS. In the following, we will provide details on team composition, working mode, and deliverables for this assignment.

2.1 Team Composition

The project assignment will be conducted in teams of approximately **5 students**. We will make no exceptions to this rule, save for *remainder teams* if the total number of course participants is not divisible by 5. Every student registered for the Examination Object in KLIPS by **Oct 27, 23:59** (see **Section 1.2**) will be assigned to a team.

2.2 Deliverables

There are two main deliverables for the project assignment. First, you must hand in a **5-page project report** (excl. figures, cover page, executive summary, references, and appendices) which is to be submitted as a .pdf file via ILIAS no later than the deadline specified in Table 1. Second, you must submit your Python code in the form of **annotated Jupyter notebooks (.ipynb)**. Additionally, you must submit a **one-page supplementary document** containing a brief description of who did what in preparing the assignment. You will be graded as a group for your project assignment (except if there is substantial reason for grade differentiation), but we do want you to reflect upon and be explicit about every team member's contribution.

Milestones: To guide your progress a little and ensure active participation, there will be two milestone deliverables during the semester. **Submission of milestones is mandatory for passing the project assignment**, but will not be graded. This is a learning from previous semesters, where it has shown that students who sign up for the course but do not plan to actively participate corrode teams' performances. Please refer to the table below for milestone deadlines.

Milestone	Description	Date/Deadline
Data Preparation	Show us that you have started working with the data	Nov 26, 23:59
Analysis & Model	Show us some progress to produce results	Jan 07, 23:59
Project Assignment	Final submission	Feb 08, 23:59

Table 2. Project Assignment Deadlines

Final Submission - Report: The project report details the project assignment, from the business problem through the data mining problem and solution, to recommendations and practical relevance. As presentation is a key component of a successful data science project, we will consider it in our evaluation. We, therefore, advise you to write your report in L^AT_EX. To facilitate the writing process, we provide a **L^AT_EX-template** on Overleaf. Using Overleaf's **documentation**, students can easily learn how to use L^AT_EX for scientific projects. Alternatively, you may use Microsoft Word using a similar typeset and line spacing. Please name the file **AA_Report_Team_<yourTeamNumber>.pdf** (e.g., **AA_Report_Team_03.pdf**).

The report should be written clearly and professionally and include the following sections:

1. **Cover page** with informative title, team number, and member names
2. **Detailed report:**
 - (a) Problem description (business goal and data mining goal)
 - (b) Data description
 - (c) Brief data preparation details (how your data was created from the raw data) and key charts. Details can be provided in an Appendix.
 - (d) Data analytics: Analytical methods applied (with sufficient detail and screenshots; use an Appendix if needed) and appropriate performance evaluation (proper choice of measures, benchmarking).
 - (e) Conclusions (advantages and limitations) and business recommendations

Final Submission - Code: Each team submits the code used to generate the findings in their report (solutions as well as figures). You must use Python (the language the course is taught in) for your project; we will not regard other programming languages. There are two options to submit your code:

1. Repository (Preferred): Submit a link to a (public) GitHub repository in which all scripts are collected. Make sure to tag a version that should be regarded for grading before the submission deadline.
2. Archive: Submit a .zip-archive containing all necessary files.

In both cases, please give the notebook files indicative names such as "01_Data_Preparation", "02_Descriptive_Analysis", etc and/or provide a README file explaining the order of execution.

Note that this course does not focus on software engineering, i.e., we do not expect you to write production-ready code. However, your submission should be sufficiently commented for us to understand it and your code will be graded as part of the final submission.

More detailed information on the project assignment (incl. the specific tasks) will be provided in a dedicated document.

3 Exam

The final exam contains questions covering material in the workshops and lectures. The exam will be of a 90-minute closed-book format and comprise multiple-choice and written sections. You will receive detailed advice on preparing for it towards the end of the course. We will offer a make-up exam this semester for those who did not pass the first exam. The registration phase for the second exam will open after the first exam. It is not possible to participate in the second exam without having been registered for the first exam (see Section 1.2). Please refer to the Class Schedule for exact dates and relevant KLIPS sign-up deadlines. **Please note that the exams take place in person and there will be no option to participate online.**

Appendix A - Course Guidelines

This section contains general rules that apply during the course. These are designed to help us provide you with a high-quality learning experience, and to make transparent what we expect from you in return. Please review these guidelines carefully. If you have any questions, please contact the instructors after looking at the paragraph on communication **(G.2)** below.

(G.1) Course Setup Classes are taught in a seminar-style format where conceptual material is discussed and related to real-world cases. Practical skills are developed in workshops. It might be difficult for you to follow the lectures if you do not study the **recommended readings** for the sessions. We encourage you to also read around the topics to further deepen your understanding of the material.

(G.2) Communication To foster a lively conversation within the class, make sure you direct your questions in the following order of priority:

- (a) There will be **weekly workshops** that serve as office hours at the same time. We will reserve time for you to ask questions of general interest during these workshops. This is the best place to ask organizational questions and clarification questions on the content of the lectures.
- (b) If your question is of general interest but cannot wait until the next workshop, or if your question is better asked in writing, post it to the **AA forum on ILIAS**. We will check the forum regularly and provide answers for everyone's benefit. In particular, all questions about drafts of your assignments must be directed to this forum, quoting from your writeup as necessary. Make sure that your questions are specific and include all relevant details. An example of a specific question is: "In the following performance requirement: [requirement text], is the number of transactions per second a good performance metric or do you know of a better alternative?" An example of a vague question is: "Attached is a file with our second group assignment. Is it good?"
- (c) If your question is personal in nature (sick leave, grades, etc.), please send it to **is3-teaching**. This email address is accessible only to the instructors and the teaching assistants of the course. In order for us to be able to optimally follow up with your request, use a subject line of the following format: *[AA] 'subject'*.
- (d) If we cannot address your problems in any of these ways, use the course email address (**is3-teaching**) to arrange a personal meeting with us. Please respect the fact that we cannot accommodate walk-in questions, no matter how brief your question may be.

(G.3) Integrity (and use of LLMs) We strictly enforce the university's policies on scholarship and plagiarism. Implicit in handing in homework, assignments, papers, and exams is that they represent your own work (or the result of sanctioned collaboration). Any exceptions must be explicitly noted. Representing someone else's work as your own is grounds for failing the course. We explicitly stress this point in light of recent advances in large language models (LLM) such as GPT. Output largely generated using LLMs does not represent your own work. This does not mean, however, that you are not allowed to use such tools - as complementary assistance, not as ghost-writers. We expect you to be explicit about the usage of tools such as ChatGPT. Specifically, you are expected to clearly indicate generated code (just like you would reference any other external source) including the specific prompt used to generate the output. For further information and guidelines, please refer to https://verwaltung.uni-koeln.de/stabsstelle02.1/content/faq/data/chatgpt/index_ger.html (in German).

(G.4) Late submissions Team and individual assignments are due at the dates and times we announce during the course. Late work is not accepted and will be treated as a fail or no-show.

(G.5) Grading Dispute Resolution Please follow the steps below in case of doubts about a grade you received:

- Sign up for a spot on the exam inspection date. As per university guidelines we will publish the dates for this six weeks in advance.
- Look at the grading scheme and your assignment again and send an email to **is3-teaching** explaining specifically where you think the grade is unjustified. Please give references to pages and specific grading comments; we have a copy of your graded assignment in our archives.

- Please respect that we cannot discuss other people's assignments, the relative merits of their writings, or their grades with you. The point of reference for us is the grading scheme and whether or not the grade you received is correct under that scheme.